TEAM REPORT

Using AI During Our Project

Steane Quantum Error-Correction Code • Development Process Report

Purpose of this report

This report briefly describes how we used AI as a supporting learning tool while exploring the Steane [[7,1,3]] quantum error-correction code. Our use of AI was limited mainly to understanding unfamiliar concepts, clarifying terminology, and improving the organization and wording of the project documentation.

How AI supported our work

AREA	LIMITED AI ASSISTANCE
1 **Concept clarification**	Helped explain the basic purpose of the Steane code and the idea of protecting one logical qubit using seven physical qubits.
2 **Terminology**	Clarified terms related to quantum error correction, such as logical and physical qubits, stabilizers, and error syndromes.
3 **Writing and organization**	Helped organize the report into a clear structure and improve readability. The project content and descriptions should reflect the work we actually carried out.

Our role

We chose the project topic and worked through the material ourselves. AI was used only when we needed a concise explanation of a concept or help presenting the information clearly. Any technical claims included in the final project should be supported by our own understanding and by work we can explain.

Responsible use of AI

> AI can be useful for learning, but its explanations may be incomplete or inaccurate. We remain responsible for understanding the material, checking facts, and ensuring that the final report accurately represents our own work. AI assistance should not be treated as evidence that a circuit, calculation, or result has been independently verified.

What we learned

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Using AI as a limited learning aid can make unfamiliar technical topics easier to approach. Asking for explanations of key terms helped us build a clearer picture of the Steane code and communicate the topic more effectively. The main value was clarification and writing support, rather than relying on AI to perform the project's technical work.

Project context

The project concerns the Steane [[7,1,3]] quantum error-correcting code. This report describes only the limited AI assistance stated above and does not claim that AI developed the code, ran simulations, generated graphs, or verified experimental results.

Scope note

> This revised version omits the earlier sections about AI-assisted code development, debugging, benchmarking, graph generation, and simulator workflows. Keep the description aligned with what was actually done during the project.